Zack Sifakis

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Research interests: *LLM reasoning, test-time compute and verification, evaluation of language models, LLM agents*

EDUCATION

PhD, Electrical and Computer Engineering *expected 05/2028*
University of Wisconsin–Madison
GPA 4.0/4.0, Advisor: Prof. Dimitris Papailiopoulos

MEng (Diploma), Electrical and Computer Engineering *10/2022*
Technical University of Crete, Greece
GPA 9.37/10 (*Distinction*), graduated 1st in class
Thesis: *Asynchronous Inference for Security Applications* (10/10)

RESEARCH EXPERIENCE

Research Assistant *Aug 2023 – present*
MLOpt Group, University of Wisconsin–Madison

- **Selecting among sampled LLM reasoning traces:** showed that lightweight hidden-state probes recover 34–40% of the headroom over majority voting on AIME across three model families (Qwen3, LFM2.5, DeepSeek-R1-Distill). Separated coverage from selection, proposed a vote-weighting diagnostic, and validated with held-out, split-stability and shuffled-label controls *(manuscript in preparation)*.
- **In-context learning in superposition:** ran experiments on GPT-3.5, Llama-3 70B and Qwen-1.5 72B measuring per-task output distributions under mixed-task prompts, establishing that pretrained LLMs execute multiple tasks at once *(ICML 2025 Spotlight)*.

Undergraduate Research Assistant *2021 – 2022*
Telecommunication Systems Institute, Technical University of Crete

- Developed distributed, asynchronous iterative inference algorithms, and built a human-presence detection system.

Research Intern *Jul 2020 – Aug 2020*
Signal Processing Lab (SPL), FORTH-ICS, Heraklion, Greece

- Single-image and video depth estimation, on 3D microscope imagery and real-time handheld phone footage.

PUBLICATIONS

Z. Xiong, Z. Cai, J. Cooper, A. Ge, V. Papageorgiou, **Z. Sifakis**, A. Giannou, Z. Lin, L. Yang, S. Agarwal, G. G. Chrysos, S. Oymak, K. Lee, D. Papailiopoulos. *Everything Everywhere All at Once: LLMs Can In-Context Learn Multiple Tasks in Superposition.* **ICML 2025 (Spotlight).**

PROJECTS

- **Bike Router** live demo – bike routing that follows rider preferences, with live bike-share data and an e-bike battery mode that guarantees a reachable charged bike (weighted Dijkstra, A*).
- **Letterboxd Friends** github – Mac app (Swift + WebKit, local Python server) showing friends' film-watching stats by year and decade; tests, CI, English and Greek versions.

SKILLS

Programming: Python, MATLAB, SQL, C/C++
ML & tools: PyTorch, Hugging Face, NumPy, Pandas, GPU clusters, Git, Linux, Docker
Languages: English (fluent), Greek (native), German (upper intermediate)

TEACHING

Teaching Assistant, ECE, UW–Madison (classes of up to 90 students) *2024 – 2025*
Neural Networks (ECE 539), Signals & Systems (ECE 330), Matrix Methods in ML (ECE 532)

HONORS, AWARDS & VOLUNTEERING

- Gerondelis Foundation Graduate Study Scholarship ($5,000)
- State Scholarships Foundation & ANEK Award for graduating 1st in class, Technical University of Crete
- Best academic performance award (2nd and 3rd year) and Academic Distinction, Technical University of Crete
- Alkiviadis Papadrakakis Foundation Scholarship for graduating top of high-school class
- *Volunteering:* IRONMAN Wisconsin, Wisconsin Film Festival, Erasmus+ exchange, earthquake relief (Greece)